Enabling the Unavailable Trip: Consumer Welfare and Willingness to Pay for Multimodal Technology

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November 8, 2026

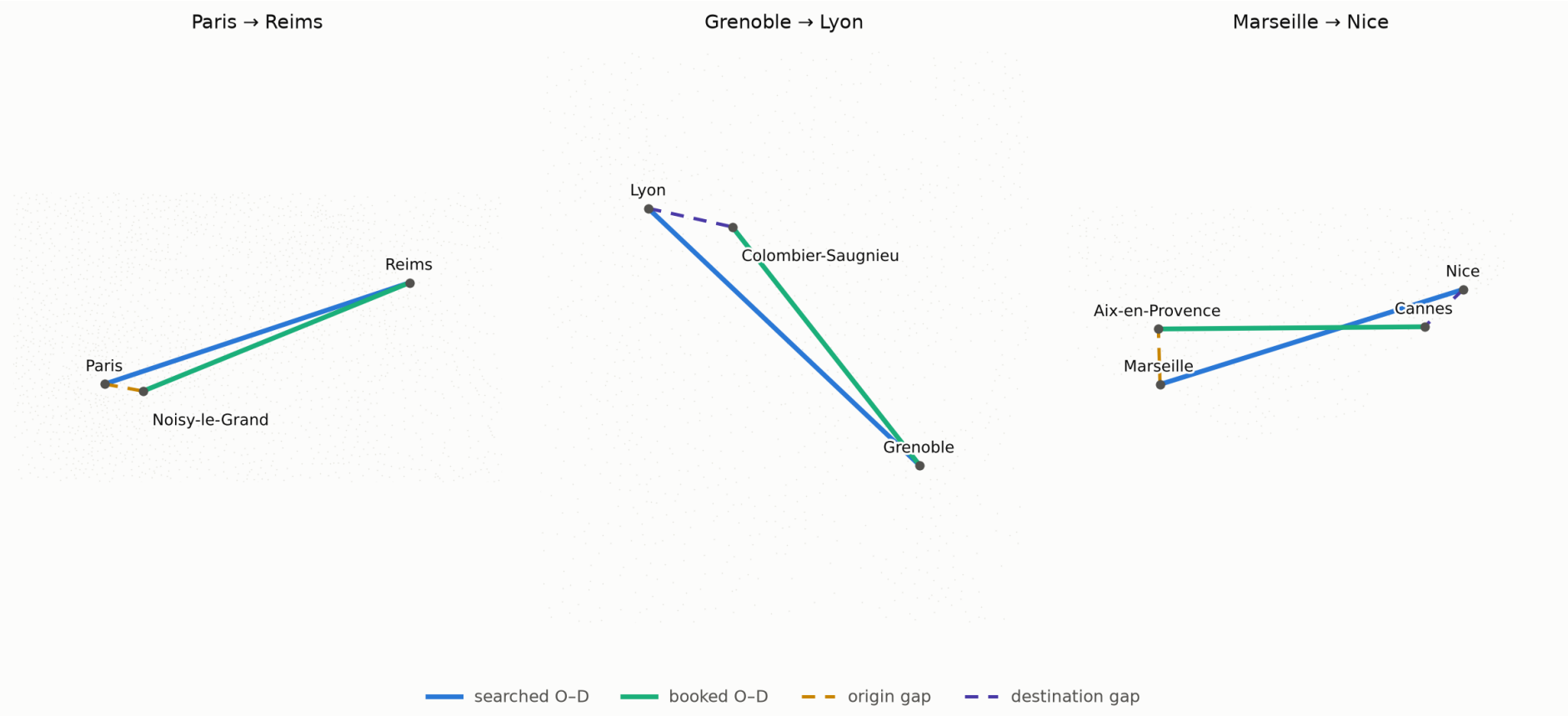
When the chosen option is not the optimal one

- In many markets, consumers cannot buy exactly the option they would most prefer.
- Frictions — search costs, limited availability, timing, coverage — push them toward a *near-match*: a substitute that differs from their ideal choice.
- The consumer then bears the cost of bridging that gap.

Yet how costly these frictions are for consumers — and how much could be gained by reducing them — remains largely unmeasured.

This paper: measure this mismatch, its cost to consumers, and the value of reducing it — in **long-distance travel**.

An example: long-distance travel



Blue: the origin–destination the traveler searched. Green: the trip actually booked. Dashed: the spatial gap the traveler must cover themselves.

Motivation

- On a long-distance platform, **about one booking in three (34%)** is for an origin and/or destination different from the one searched.
- Policy is pushing toward an integrated, multimodal transport sector (European Commission 2017, 2023; European Parliament 2026) — digital platforms can support it by letting travelers **combine modes within a single trip**, making otherwise-unavailable itineraries feasible.

Demand costs and supply frictions

On the demand side, a traveler who cannot book the searched trip faces search and coordination costs — and the cost of covering the spatial gap themselves.

On the supply side, two frictions generate these mismatches:

- **Coverage:** the searched origin–destination (OD) is not served.
- **Capacity / timing:** a trip exists, but none matches the searched OD when the traveler searches.

Multimodal integration can relax both — more OD pairs feasible (extensive margin) and more options on served routes (intensive margin).

This paper

We study **BlaBlaCar**, the largest long-distance carpooling platform, which also operates its own scheduled buses (*BlaBlaBus*) — two intercity modes on a single interface, in France.

Three questions:

- How large are the frictions that keep passengers from finding suitable itineraries? [preliminary evidence]
- How much do these frictions affect travel choices and welfare? [structural demand model]
- To what extent can multimodal integration improve consumer outcomes? [counterfactuals]

Preliminary results

- **Spatial mismatches are common — and not just missing supply.**
  ≈34% of bookings differ from the searched OD; gaps average ≈50 km, while only a small share of searches is unserved.

- **Mismatches are costly, and consumers would pay to remove them.**
  Disutility is driven by large origin/destination gaps — consumers are largely indifferent to minor ones; WTP ≈ EUR 14.1 per 100 km, EUR 6.95 for the average 49.3 km gap (about a third of a fare), and higher for older and privacy-sensitive travelers.

- **Closing the gap raises welfare, expands the market, and can be priced.**
  +EUR 0.57 per searcher (EUR 5.6 per booking; ≈EUR 10.4M, a quarter of a fare, no new capacity); booking rate rises 10.75%→11.6%, mostly carpool; break-even surcharge EUR 5.12, about 25% of the fare.

Related literature

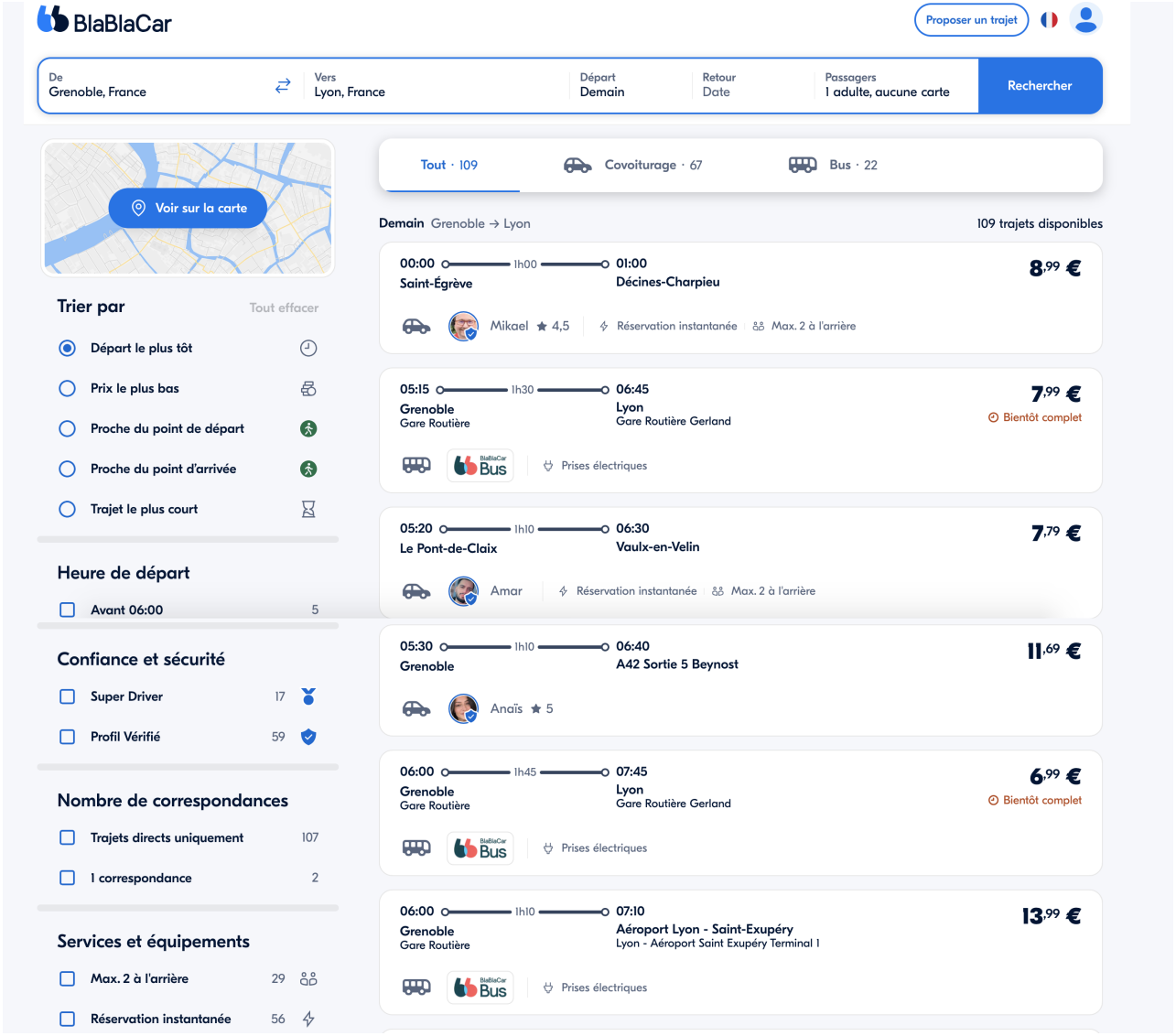
1. **Market and matching frictions in transportation.**
   Lagos (2000); Fréchette et al. (2019); Buchholz (2022); Bian (2018); Gaineddenova (2022); Castillo (2025)

2. **Digital transportation platforms and multimodal services.**
   Buchholz et al. (2020); Hall et al. (2018); Castillo et al. (2025); Delhomme & Gheorghiu (2016); Monchambert (2020); Farajallah et al. (2019); Yeung & Zhu (2022); Astier et al. (2023)

Our contribution: we identify and *quantify* the consumer cost of searched-vs-booked mismatches and the value of integrating modes — not measured before, for want of linked search–booking data and with demand models too coarse to price the gap.

Institutional background & data

BlaBlaCar platform



Data

Proprietary BlaBlaCar data — 100 most popular routes in France, Nov 2021–Nov 2022.

- **Searches** (one row per query): searched OD, date, displayed choice set and prices, clicks, and demographics (gender, age group).
- **Bookings** (carpool and bus files): booked OD, actual pick-up/drop-off, prices, seats, and trip distance and duration.
- **Supply**: bus inventory (services, capacity, stops) and cars offered per route–date; cars and buses at each location (endpoint connectivity).
- **Additional**: municipality coordinates (to measure spatial gaps) and daily fuel prices (price instruments).

≈18.3M searches, ≈1.16M bookings.

▸ Descriptive statistics

Preliminary evidence

Preliminary evidence: what we document

1. **Magnitude and heterogeneity** of origin–destination mismatches.
2. **Determinants of the frictions** — the nature of a mismatch, across modes and margins.
3. **Willingness to multimode** — do travelers accept, and value, alternative itineraries?

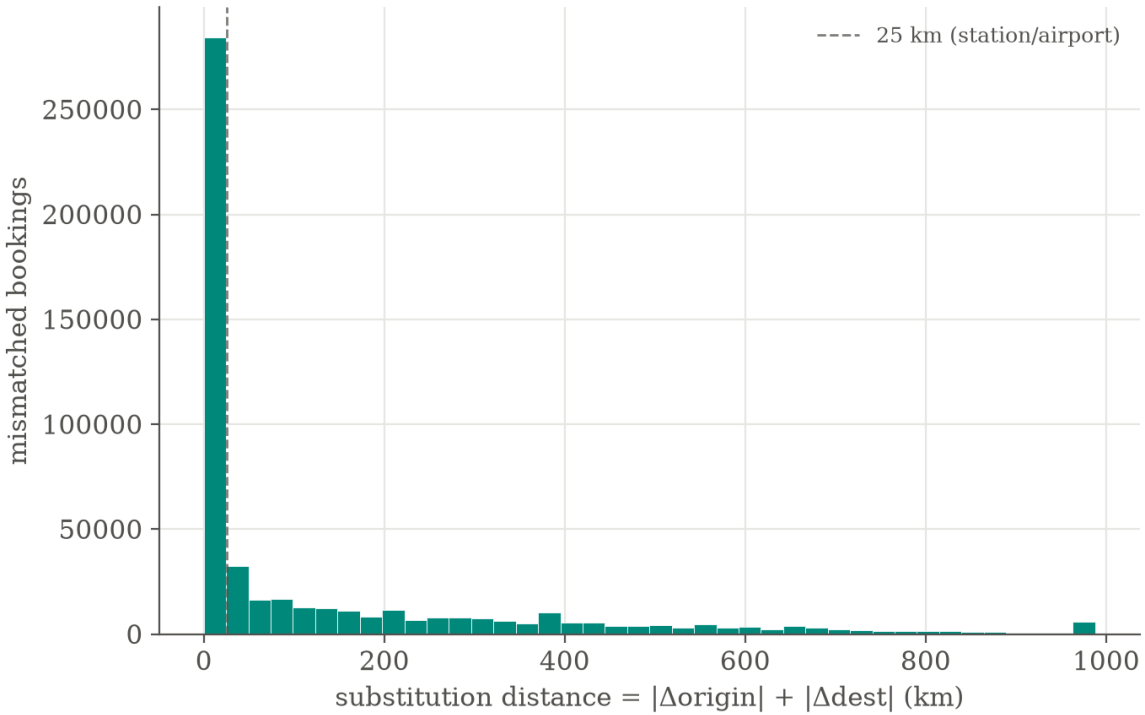
One booking in three is a near-match — and far more for carpool

	Total	Bus	Carpool
Number of bookings	1,163,897	629,626	534,271
Search–booking match (%)	66.0	82.2	46.9
Origin gap only (%)	15.1	5.8	26.1
Destination gap only (%)	16.1	11.9	21.0
Origin and destination gaps (%)	2.9	0.2	6.1
Trip distance (km)	210.0	223.0	200.0
Origin gap (km)	12.5	133.7	10.2
Destination gap (km)	10.7	5.8	13.0
Origin + destination gap (km)	46.2	23.2	46.9

Distance rows are medians. 34% of bookings are near-matches: 53% of carpool vs. 18% of bus.

▸ Mismatch maps

Most mismatches are small — but a long tail is large



CDF of the origin + destination gap; strongly right-skewed — about 55% of near-matches are within 25 km, the rest substantially larger.

▸ Mismatch maps

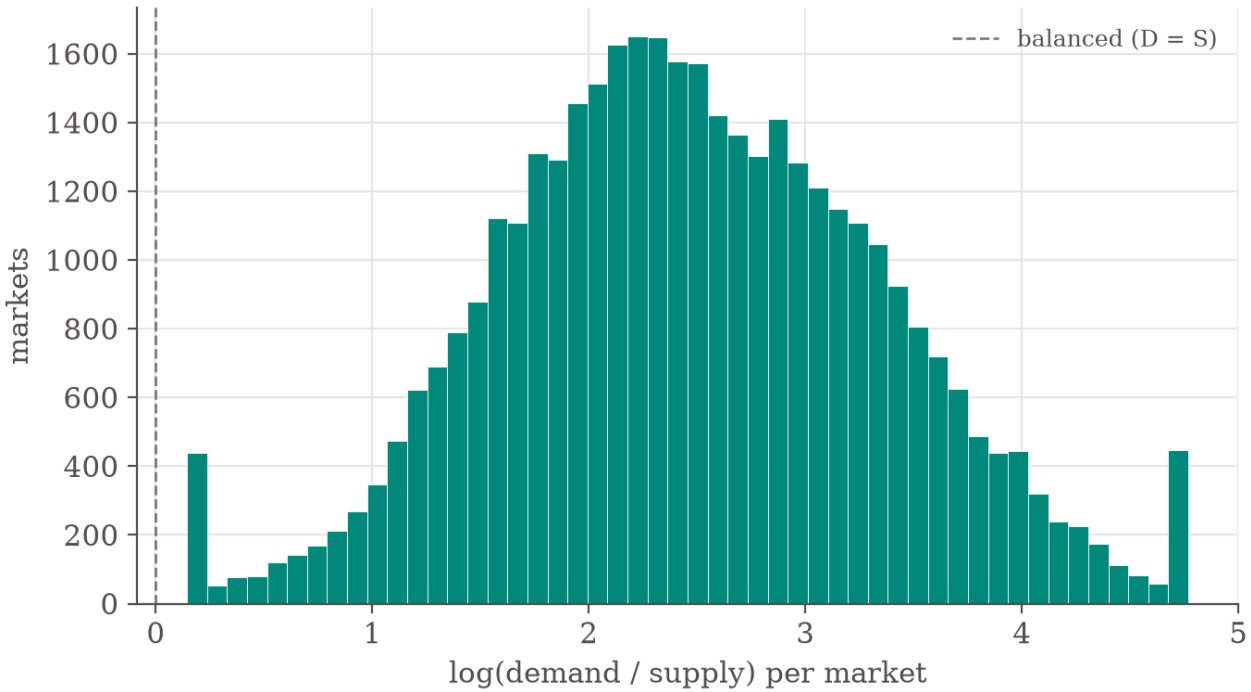
Options usually exist — the friction is matching, not coverage

	Underserved no supply	Congestion supply, no option	Other option existed
A. Near-match bookings (booked OD $\neq$ searched)			
Carpool	8.6	0.9	90.6
Bus	49.7	2.2	48.0
B. Searches without a booking			
Carpool	8.1	11.0	80.9
Bus	24.6	5.3	70.1

All values in %. Denominators: near-matched bookings (A; carpool 283,901, bus 111,913) and all searches (B; 20,621,323 per mode). Across both margins "option existed" dominates — buses aside, where missing coverage explains about half of near-matches and a quarter of unbooked searches.

▸ Displayed vs. offered

Displayed options fall short of demand in nearly every market



Distribution of $\log(\text{demand}/\text{supply})$ across markets (searches vs. displayed results); mass almost entirely above zero. What a traveler sees is the supply *at the time of search*, not total supply on the route.

Willingness to multimode: specification

$$P(Y = 1 \mid \cdot) = G\big(\beta_1\, Conn_{o,d} + \beta_2\, Conn_{o,d} \times Unserved_{o,d} + \beta_3 X'\big)$$

- $Conn_{o,d}$: cars + buses touching the searched origin and destination (leave-one-out).
- X': displayed price and displayed direct supply; SE clustered on origin.

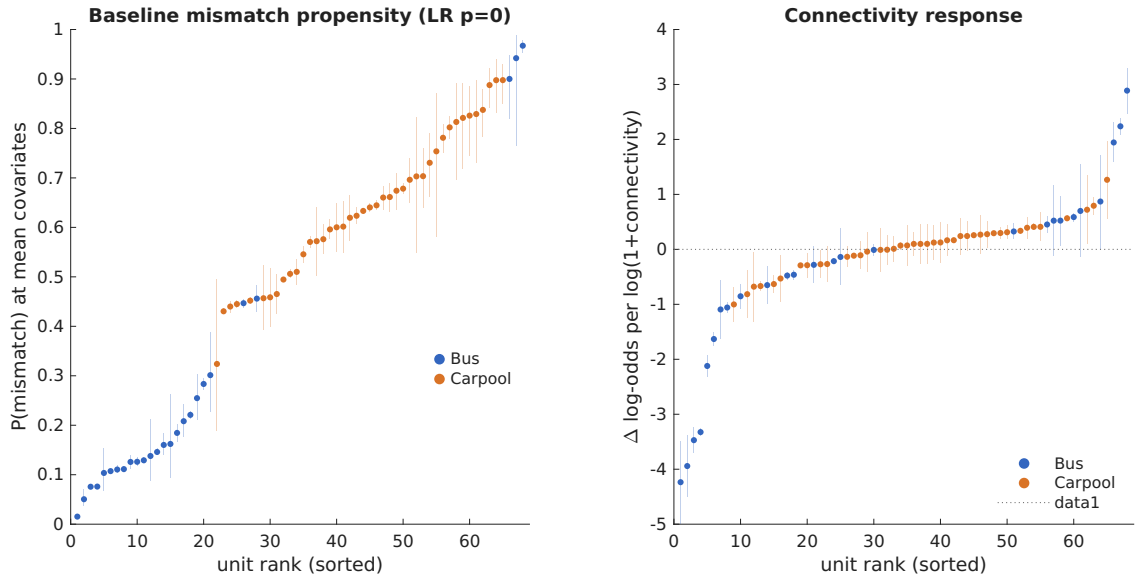
Three outcomes:

(A) **Conversion** — book, over all searches.

(B) **Any near-match** — booked OD $\neq$ searched, conditional on booking.

(C) **Severe near-match** — gap above the median.

Propensity to multimode varies widely — and rises with connectivity for carpool



Each point = origin×mode unit ($\geq$50 bookings). *Left*: baseline near-match probability (higher for carpool). *Right*: response to endpoint connectivity (mostly ≥ 0); common response rejected ($p \approx 0$). In the regressions, connectivity raises near-match bookings significantly for carpool, not for bus.

▸ Connectivity estimates

Structural model

Modeling choices

Why a structural model? To recover price sensitivity and the disutility of the gap, and to quantify the welfare value of integration — objects the reduced form cannot deliver.

- **Static discrete choice**: consumers decide at the time of search, over the trips available then; we abstract from purchasing-horizon dynamics.
- **Each search is an independent choice** (no consumer panel).
- **Choice set**: trips aggregated to mode $\times$ 6-hour window $\Rightarrow$ 9 alternatives (8 inside + outside).
- **Outside option**: own car, train, other operators, or no trip.
- **Unobserved tastes**: restricted factor structure — platform, carpool, and departure-window shocks $(\sigma_P, \sigma_C, \sigma_W)$.
- **Estimation sample**: most-booked routes, up to 24 dates each (≈1,458 markets, 89 routes, 13 months, ≈169k bookings).

The spatial gap G_{jm}

$$G_{jm} = \underbrace{g^o_{jm}}_{\text{origin gap}} + \underbrace{g^d_{jm}}_{\text{destination gap}}, \qquad G_{jm} = 0 \text{ for an exact match} \tag{1}$$

- g^o_{jm}, g^d_{jm}: great-circle distance between the *searched* and *booked* origin (resp. destination), averaged over the bookings of trip j, per 100 km.
- Main measure: the sum $g^o + g^d$. Positive for a near-match; the displacement the traveler must cover themselves.
- Robustness: origin/destination gap shares, split into *minor* (0.5–25 km) and *severe* (>25 km).

Demand — individual utility

$$U_{ijm} = \underbrace{\delta_{jm}}_{\text{baseline utility}} + \underbrace{\pi^{p}\, d_i\, p_{jm} + \pi^{x}\, d_i\, x_{jm} + \varepsilon_{ijm}}_{\text{deviation from baseline}} \tag{2}$$

- i indexes the consumer; d_i are demographics (age, gender).
- p_{jm} price; x_{jm} trip characteristics (bus vs. carpool, route distance, departure window).
- π^{p}, π^{x}: how price sensitivity and mode preferences vary across demographics.
- $\varepsilon_{ijm} \sim MVN(0, \Omega)$: unobserved taste, correlated across trips (factor structure).
- Outside option normalized: $U_{i0m} = \varepsilon_{i0m}$ (own car, train, other operators, no trip).

Demand — baseline utility δ_{jm}

$$\delta_{jm} \;=\; \alpha\, p_{jm} \;+\; \beta\, x_{jm} \;+\; \gamma\, G_{jm} \;+\; \xi_{jm} \tag{3}$$

- α: common price sensitivity; β: common taste for trip characteristics.
- γ: disutility of the searched↔booked gap — *the object of interest*.
- ξ_{jm}: unobserved trip quality (amenities, driver attributes) — source of price endogeneity.

Estimation and identification

Two-stage, individual-level estimation (Goolsbee–Petrin, 2004) — a micro-BLP using consumer choices directly.

First stage (micro):

- SML with trip–market FE δ_{jm}; recovers heterogeneity π^p, π^x and covariance $\sigma_P, \sigma_C, \sigma_W$.

Second stage (market):

- Project δ_{jm} on price, characteristics, gap (2SLS) $\Rightarrow \alpha, \beta, \gamma$.

- **Flexible substitution** via a factor structure on tastes:

$$\varepsilon_{ijm} = \sigma_P \, \nu_i^P + \sigma_C \, \nu_{i,c(j)}^C + \sigma_W \, \nu_{i,w(j)}^W + \eta_{ijm}$$

Identification:

- **Variation in the searched↔booked gap** across displayed trips identifies γ
  a near-match in m is an exact-match posted for another market m'; the algorithm, not the driver, displays it in m
- **Cost-per-seat shifters** (fuel $\times$ distance, tolls, labor) identify price sensitivity α
- **Demographic variation in bookings** identifies the heterogeneity π^p, π^x

Results

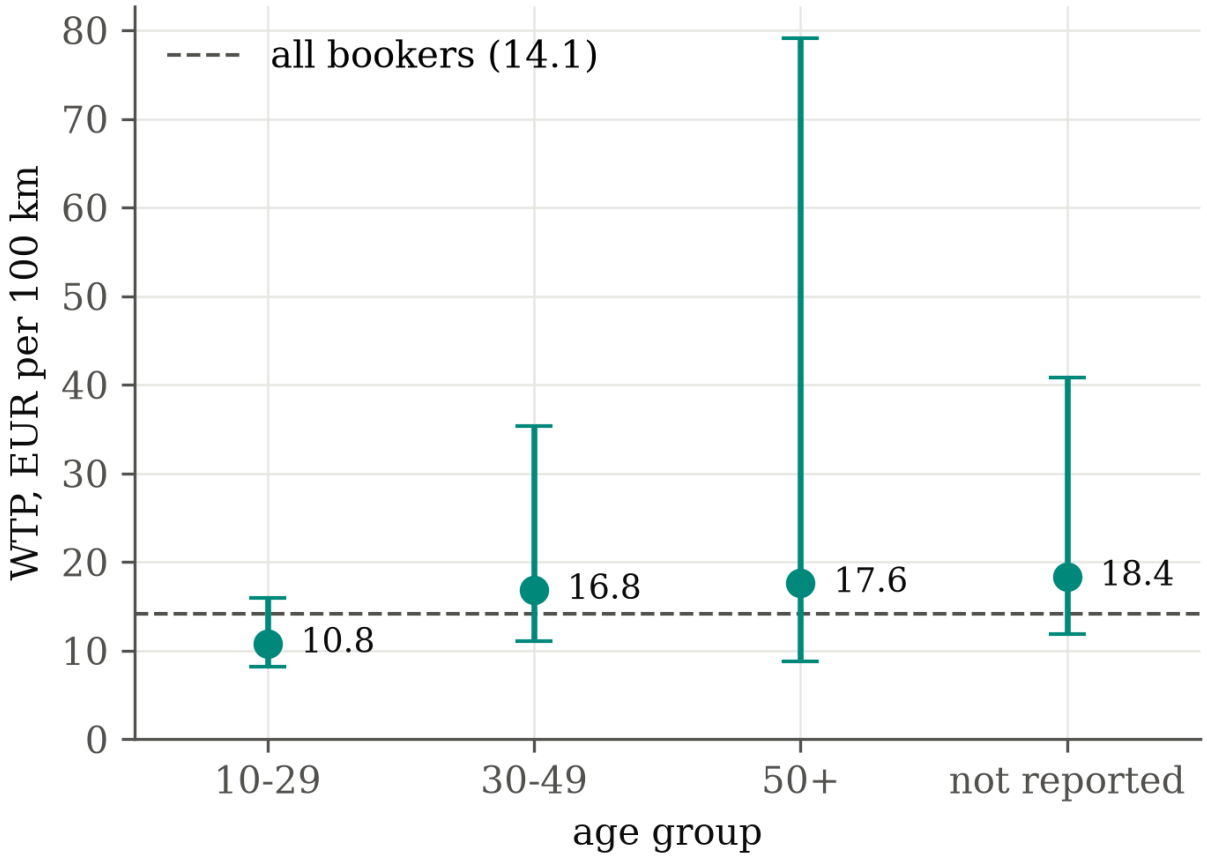
Structural results

	Estimate
Panel A. Demographic heterogeneity — first stage (SML)	
Price × age 30–49	0.007^{***}
Price × age 50+	0.007
Price × unreported (privacy)	0.008^{***}
Carpool × female	0.164^{***}
σ_P (platform)	0.746
σ_C (carpool)	0.633
σ_W (departure window)	0.531
Panel B. Common preferences — second stage (2SLS)	
Price α	-0.018^{***}
Gap γ (per 100 km)	-0.199^{***}
Bus	-0.003
Observations (stage 1 / stage 2)	169,052 / 9,533
First-stage F	29.1

Stage 1 col. (3), stage 2 col. (4). Price interactions are deviations from the base group (age 10–29, male); a positive sign = less price-sensitive.
$^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$. Full tables and standard errors in the appendix.

▸ Full estimates

Willingness to pay



WTP in EUR per 100 km of gap. Average 14.1 EUR/100 km; 6.95 EUR for the average 49.3 km gap (≈ a third of a fare). Age 10–29 pay below 11 EUR; older groups 56–63% more; privacy-sensitive (unreported) 18.4 EUR. Point estimates with 5–95% parametric-bootstrap range.

▸ How WTP is computed

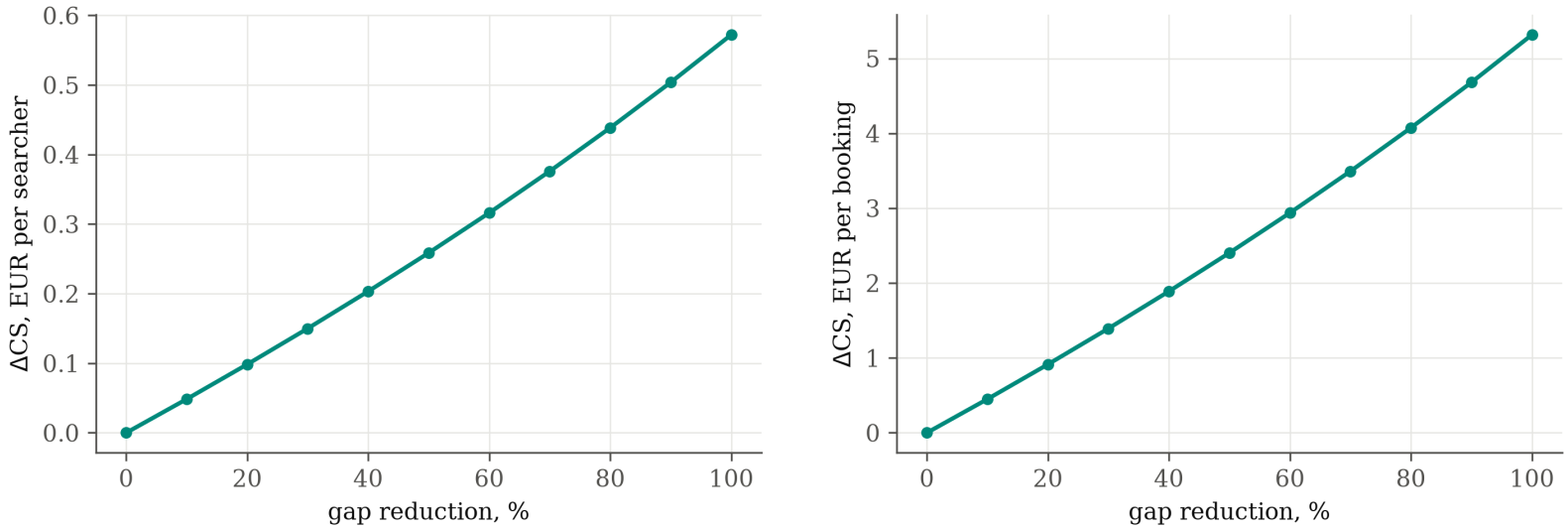
Counterfactual welfare analysis

Counterfactual analysis

- Use the estimated model to value **multimodal technology**: progressively close the searched↔booked gap (observed → full elimination), holding prices and supply fixed (demand-side, partial equilibrium).

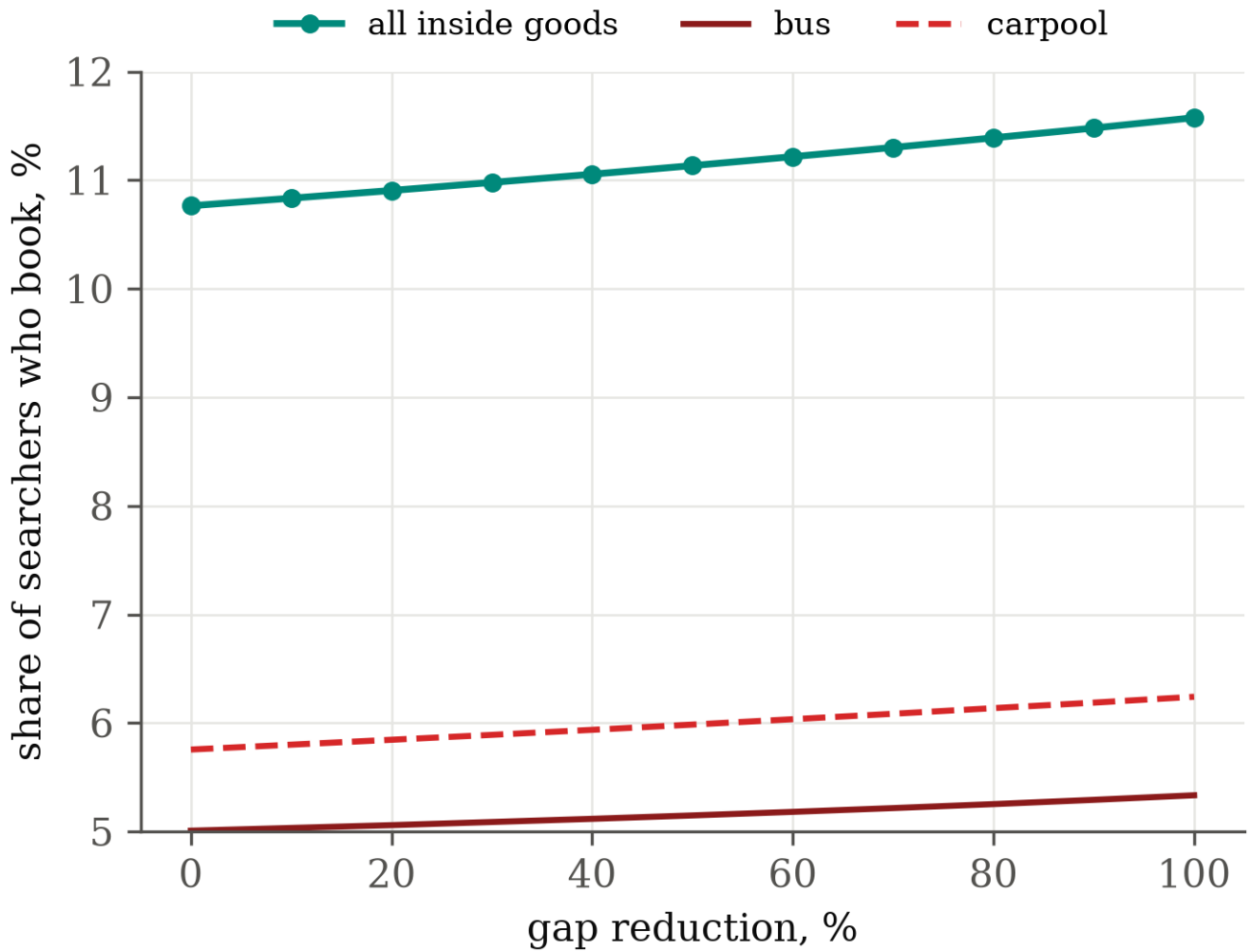
  (a) **Consumer surplus** — compensating variation as the gap shrinks, per searcher and per booking.
  (b) **Extensive margin** — how the share of searchers who book rises, and for which mode.
  (c) **Pricing the technology** — a uniform surcharge: how much before the gains vanish.

(a) Closing the gap raises consumer surplus



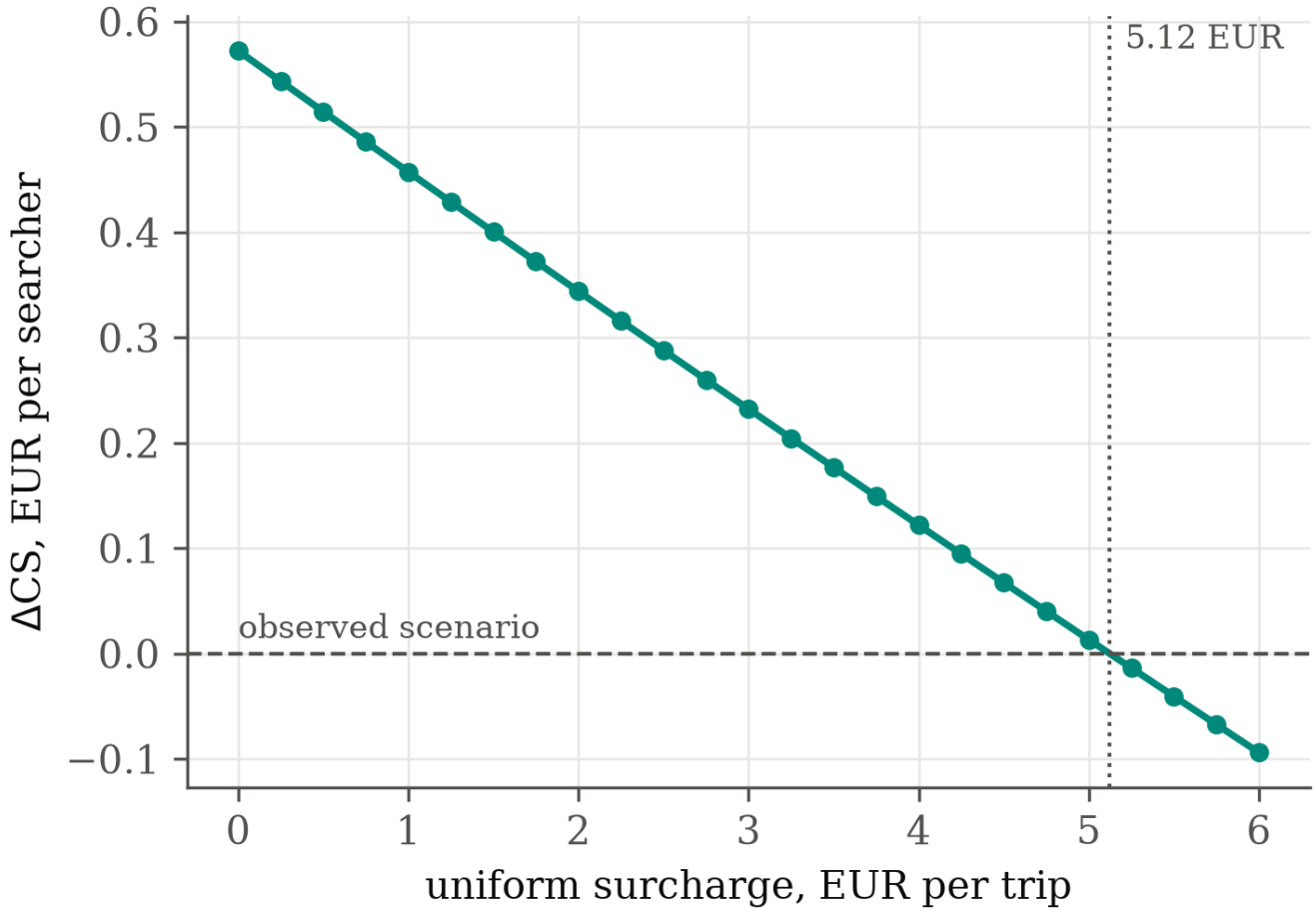
Change in consumer surplus as the gap is reduced (0% = observed). Full elimination: $+0.57$ EUR per searcher (left) and $+5.6$ EUR per booking (right) — ≈10.4M EUR in aggregate, about a quarter of a fare, with no new capacity.

(b) ...and expands the market, mostly through carpool



Share of searchers who book as the gap is reduced. Overall $10.75\% \to 11.6\%$ $(+8\%)$; carpool $5.75\% \to 6.25\%$ vs. bus $5.0\% \to 5.33\%$.

(c) The platform can price it — up to ≈5 EUR per trip



Consumer surplus per searcher under full gap elimination against a uniform surcharge. Break-even at 5.12 EUR per trip (≈ a quarter of a fare, near the current commission); below it the average consumer is better off.

Conclusion

Conclusion

- We study the value of **multimodal technology** on a long-distance platform, using the spatial gap between searched and booked trips.
- Spatial mismatches are common (34% of bookings) and **costly**: consumers would pay 6.95 EUR to close the average gap.
- Eliminating the gap raises welfare and expands the market:
  - +10.4M EUR in consumer surplus, with no new capacity.
  - Booking rate up 8%, mostly through carpooling.
- The platform could **price** the technology (up to ≈5 EUR per trip) and still leave the average consumer better off.

A broader research agenda

- This paper is the **first step**: it documents and prices the consumer cost of searched-vs-booked mismatches, and the demand-side value of integration.
- **Next**: a structural matching model of the two-sided platform (ride requests, bookings, BlaBlaBus) — entry, matching, and platform pricing.
- **Methodology**: a companion paper developing a method to estimate the gap when it is *not directly observed*.
- **Beyond**: the same mismatch lens — optimal vs. chosen option under search frictions — applies to other markets.

Thank you!

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Appendix: descriptive statistics — searches, mode choice, demographics

	Mean	Std. Dev.
Number of searches	18,259,760	
Number of bookings	1,163,897	
Bus (%)	3.4	18.2
Carpool (%)	2.9	16.9
Outside option (%)	93.6	24.4
Women (%)	49.1	50.0
Age 10–19 (%)	9.2	28.9
Age 20–29 (%)	39.3	48.8
Age 30–39 (%)	12.3	32.8
Age 40–49 (%)	5.8	23.4
Age 50–59 (%)	3.3	17.9
Age 60+ (%)	1.3	11.5
Age unreported (%)	28.9	45.3

Shares relative to searches; demographics from the booking data.

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Appendix: descriptive statistics — route–day–time of day level

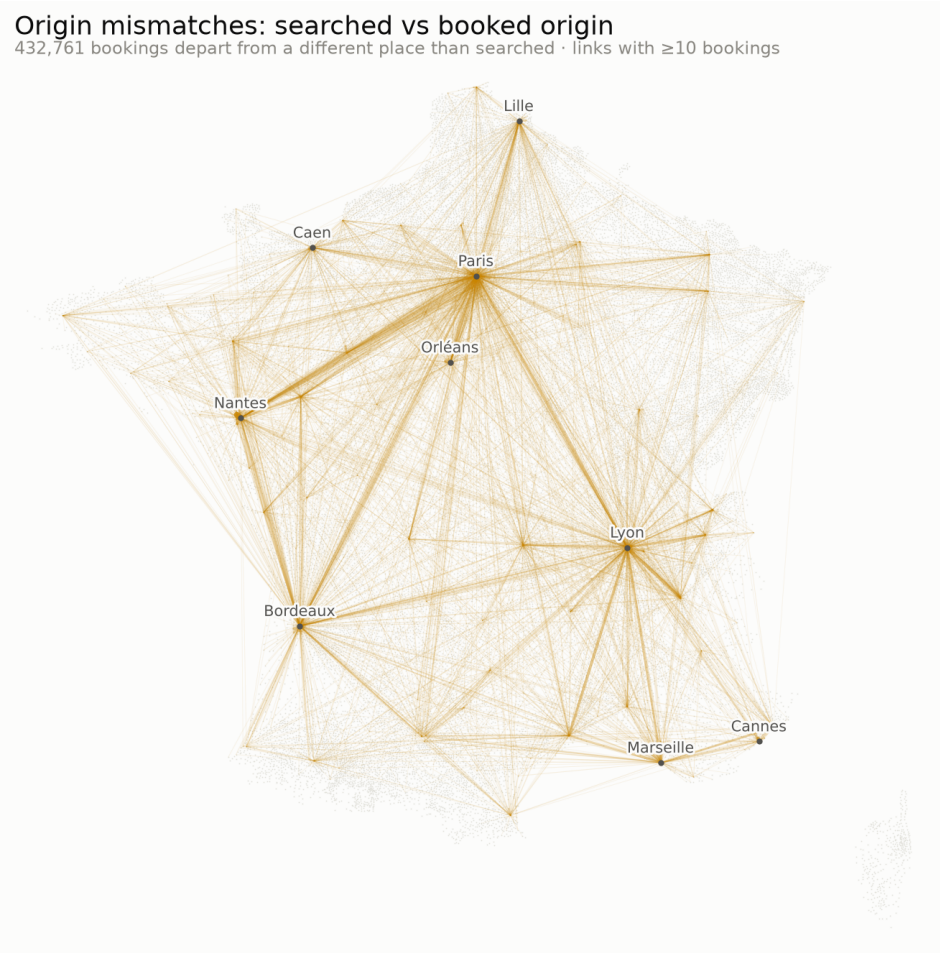
	Mean	Std. Dev.
Number of observations	79,970	
Price bus (EUR)	20.34	21.77
Price carpool (EUR)	24.39	16.51
Driver fee (EUR)	18.13	12.34
Commission (EUR)	5.32	3.57
VAT (EUR)	0.93	0.66
Monday (%)	14.8	35.5
Tuesday (%)	11.5	31.9
Wednesday (%)	11.8	32.3
Thursday (%)	14.1	34.8
Friday (%)	16.6	37.2
Saturday (%)	15.4	36.0
Sunday (%)	15.8	36.5
Departure 0h–6h (%)	17.1	37.6
Departure 6h–12h (%)	31.5	46.4
Departure 12h–18h (%)	30.8	46.2
Departure 18h–24h (%)	20.7	40.5
Trip distance (km)	273.7	176.4
Trip duration (hours)	3.13	1.87
Total distance gap (km)	49.3	92.9

Unit: searched route × date × six-hour window. Carpool price = driver fee + commission + VAT.

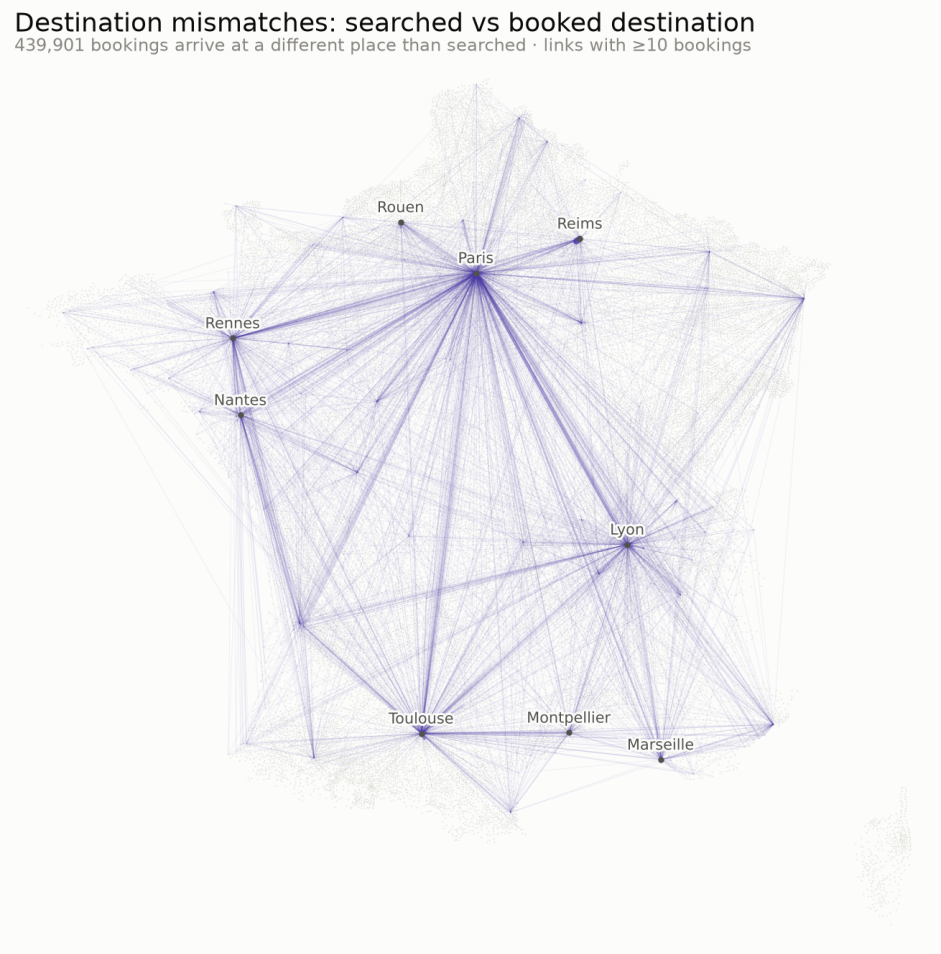
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Appendix: geographic distribution of mismatches

Origin mismatches



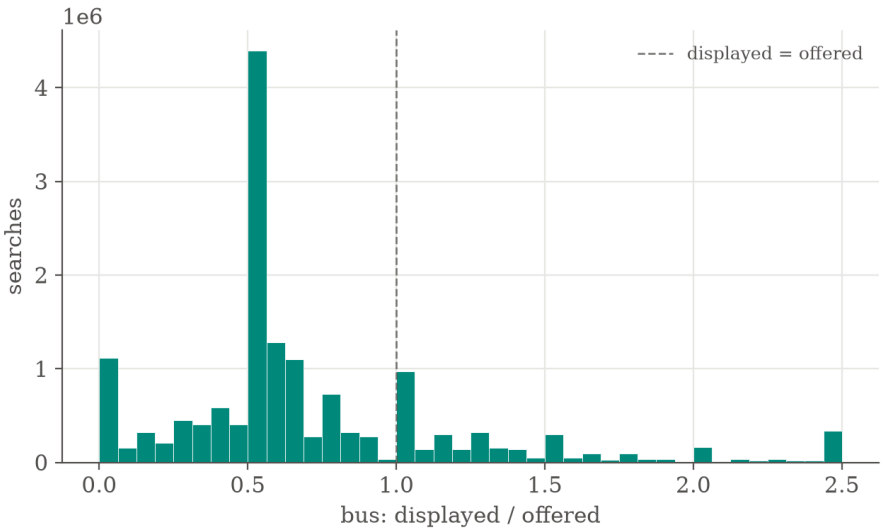
Destination mismatches

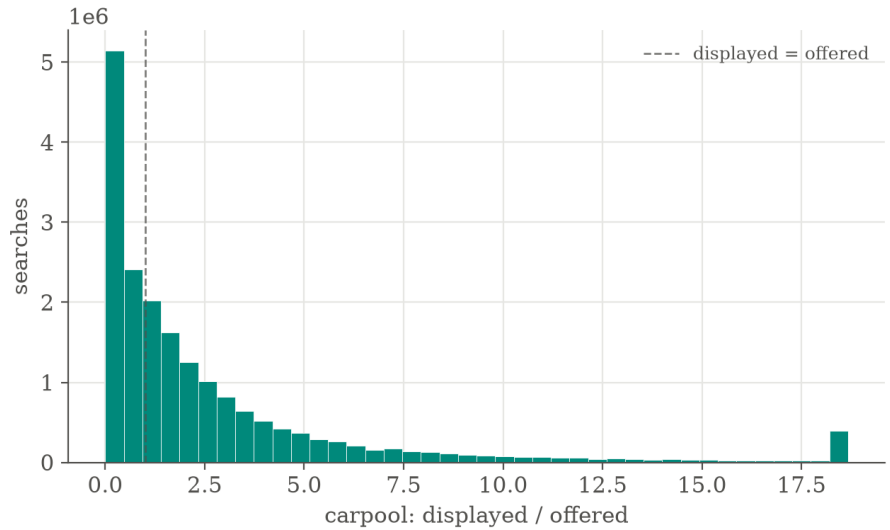


Node size = number of near-match bookings at that location; near-matches occur throughout the network, concentrated around large cities and in the south.

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Appendix: displayed vs. offered supply, by mode





Bus (left), carpool (right). Median displayed/offered ratio 0.56 (bus) and 1.37 (carpool); measured differently across modes, so not directly comparable. Mass at zero in both.

◂ Back

Appendix: endpoint connectivity, conversion, and near-matches

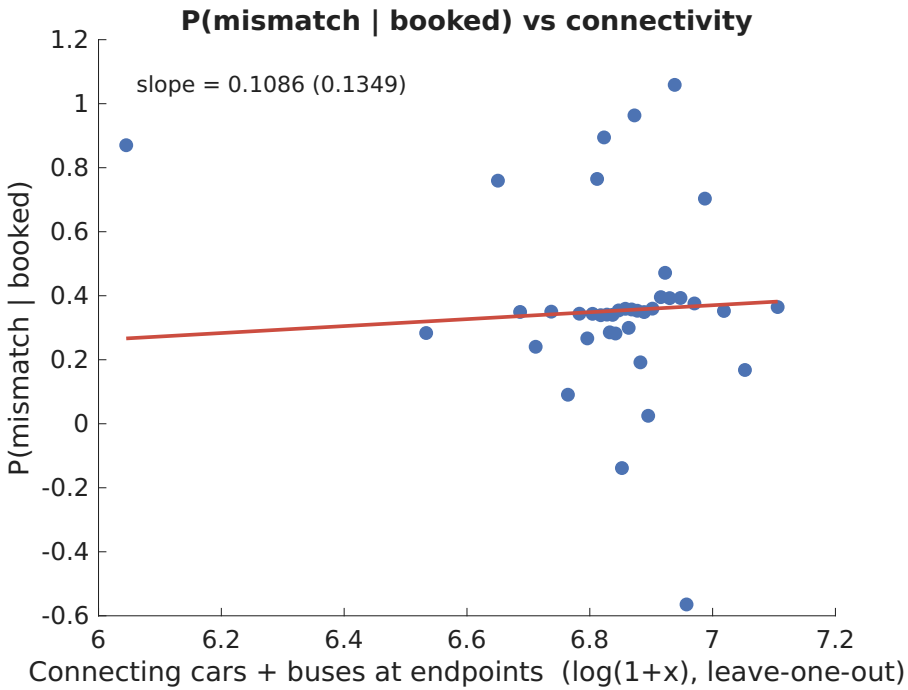
	Probit (main)			Logit		
	Pooled	Bus	Car	Pooled	Bus	Car
Panel A. Conversion: Pr(book), all searches						
All searches	-0.096^{***}	-0.075^{*}	-0.023	-0.182^{***}	-0.138^{*}	-0.049
Carpool	-0.216^{***}	-0.411^{***}	$+0.187^{**}$	-0.438^{***}	-0.854^{***}	$+0.380^{**}$
Bus	$+0.061$	$+0.391^{***}$	-0.306^{**}	$+0.137$	$+0.865^{***}$	-0.679^{**}
Panel B. Any near-match: Pr(near-match | booked)						
All bookings	$+0.131$	$+0.120$	$+0.018$	$+0.223$	$+0.182$	$+0.049$
Carpool	$+0.132^{***}$	$+0.018$	$+0.114$	$+0.210^{***}$	$+0.028$	$+0.183$
Bus	$+0.087$	$+0.267$	-0.131	$+0.158$	$+0.502$	-0.244
Panel C. Severe near-match (vs. matched bookings)						
All bookings	$+0.131^{***}$	$+0.085$	$+0.046$	$+0.255^{***}$	$+0.144$	$+0.110$
Carpool	$+0.234^{***}$	$+0.115$	$+0.116$	$+0.393^{***}$	$+0.191$	$+0.193$
Bus	-0.072	-0.198	$+0.095$	-0.140	-0.377	$+0.174$

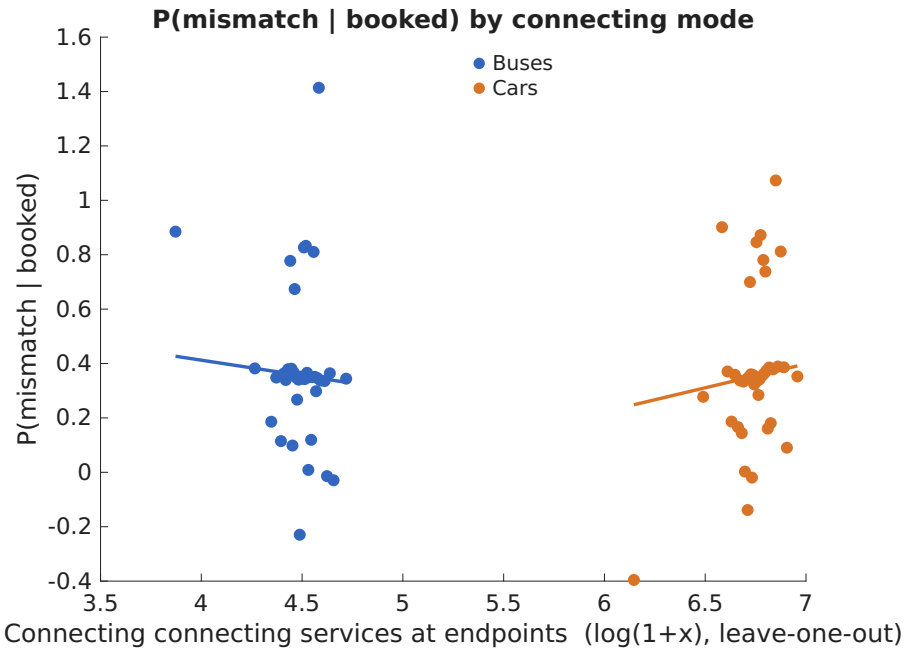
Each cell: coefficient on endpoint connectivity from a separate regression.

$^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

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Appendix: connectivity dose–response (within OD and day)





Probability a booking is a near-match vs. endpoint connectivity: pooled (left) and by connecting mode (right).

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Appendix: first-stage estimates (1/2) — price & bus interactions

	(1)	(2)	(3)
Price × demographics			
Price × age 30–49	−0.001	0.000	0.007
	(0.002)	(0.000)	(0.001)
Price × age 50+	0.002	−0.010	0.007
	(0.004)	(0.005)	(0.006)
Price × age unreported	0.020	0.022	0.008
	(0.001)	(0.001)	(0.001)
Bus × demographics			
Bus × female		−0.754	0.003
		(0.169)	(0.002)
Bus × age 30–49			0.015
			(0.003)
Bus × age 50+			−0.051
			(0.006)
Bus × age unreported			0.250
			(0.017)

SML conditional on booking. Base: age 10–29, male. Col. (3) preferred. SE in parentheses.

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Appendix: first-stage estimates (2/2) — carpool & taste heterogeneity

	(1)	(2)	(3)
Carpool × demographics			
Carpool × female			0.164
			(0.031)
Carpool × age 30–49			0.089
			(0.030)
Carpool × age 50+			0.024
			(0.006)
Carpool × age unreported			−0.867
			(0.106)
Taste heterogeneity (std. dev.)			
Platform shock	0.858	1.692	0.746
	(0.014)	(0.008)	(0.011)
Carpool shock	0.181	0.051	0.633
	(0.016)	(0.004)	(0.019)
Departure-window shock	0.712	0.534	0.531
	(0.015)	(0.017)	(0.012)
Log-likelihood	−266,855.7	−264,303.5	−257,696.3
Observations	169,052	169,052	169,052

Taste heterogeneity: std. dev. of normal shocks (platform, carpool, departure window). Col. (3) preferred.

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Appendix: second-stage estimates (1/2) — price & gaps

	OLS		2SLS (IV11)			
	(1)	(2)	(3)	(4)	(5)	(6)
Price × bus	−0.008		−0.022			
	(0.001)		(0.004)			
Price × carpool	−0.008		−0.025			
	(0.001)		(0.004)			
Price		−0.008		−0.018	−0.023	−0.036
		(0.001)		(0.004)	(0.004)	(0.005)
Gap sum (100 km)	−0.198	−0.198	−0.202	−0.199		
	(0.007)	(0.007)	(0.008)	(0.008)		
Origin gap, minor					−0.052	0.035
					(0.040)	(0.047)
Origin gap, severe					−0.732	−0.544
					(0.035)	(0.053)
Destination gap, minor					−0.004	0.064
					(0.039)	(0.047)
Destination gap, severe					−0.749	−0.498
					(0.035)	(0.058)

Stage-2 regressions of $\hat{\delta}$. Cols. (1)–(2) OLS, (3)–(6) 2SLS. Col. (4) preferred; (5)–(6) split the gap into minor/severe. SE in parentheses.

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Appendix: second-stage estimates (2/2) — controls & fit

	OLS		2SLS (IV11)			
	(1)	(2)	(3)	(4)	(5)	(6)
Log searches						−0.009
						(0.060)
Log offers						0.211
						(0.022)
Log results displayed						0.299
						(0.021)
Departure 0–6h	−0.931	−0.931	−0.933	−0.932	−0.939	−0.996
	(0.019)	(0.019)	(0.019)	(0.019)	(0.019)	(0.022)
Departure 12–18h	0.105	0.105	0.089	0.094	0.088	0.071
	(0.016)	(0.016)	(0.017)	(0.017)	(0.017)	(0.018)
Departure 18–24h	−0.512	−0.512	−0.493	−0.496	−0.502	−0.539
	(0.015)	(0.015)	(0.017)	(0.017)	(0.017)	(0.020)
Bus	−0.008	−0.023	−0.077	−0.003	−0.053	0.892
	(0.025)	(0.013)	(0.030)	(0.014)	(0.018)	(0.043)
Constant	−3.182	−3.176	−2.964	−3.046	−2.775	−4.419
	(0.176)	(0.175)	(0.158)	(0.162)	(0.133)	(0.361)
Observations	9,533	9,533	9,533	9,533	9,533	8,630
F-stat			70.4 / 72.8	29.1	29.2	18.9

Network controls (col. 6), departure dummies (base 6–12h), bus dummy, and fit. First-stage F on the excluded instruments.

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Appendix: price elasticities and diversion ratios

	Bus	Carpool	Outside option
Panel A: Price elasticities			
Bus	−1.095	0.170	
Carpool	0.154	−1.059	
Outside option	0.047	0.056	
Panel B: Diversion ratios			
Bus	–	0.167	0.833
Carpool	0.134	–	0.866

Mode-level aggregates from stage-1 col. (3) and the col. (6) price coefficient (−0.036).

Observed shares: bus 0.046, carpool 0.055, outside 0.899.

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Appendix: computing willingness to pay

WTP is the money a consumer trades off against the gap — the marginal rate of substitution between gap and price:

$$\mathrm{WTP}_i = \frac{\gamma}{\alpha_i}\,\Delta G, \qquad \alpha_i = \alpha + \pi^p d_i$$

- γ: gap disutility (second stage, spec. 4); α_i: heterogeneous price sensitivity (both stages).
- Both are negative, so $\gamma/\alpha_i > 0$ — euros a consumer would pay to remove a unit of gap.
- Expressed per 100 km, then evaluated at the average gap $\Delta G = 49.3$ km $\Rightarrow$ 6.95 EUR.
- Heterogeneity enters through α_i: younger, more price-sensitive travelers pay less; privacy-sensitive pay most.
- Confidence bands: parametric bootstrap (10,000 draws).

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